

Placement Portal Application (2026)

Java Full Stack Learning Project (100% Free Local Development)

Objective

Build an industry-style Placement Portal Application while learning modern Java Full Stack development.

The project will be developed completely on a local machine using free tools and technologies.

The project focuses on:

- Java 25
- Spring Boot
- Spring Data JPA
- Hibernate
- Spring Security
- JWT Authentication
- MySQL/PostgreSQL
- React + Vite
- Tailwind CSS
- Redis
- RabbitMQ
- Docker
- Swagger/OpenAPI
- JUnit & Mockito
- Git & GitHub
- GitHub Actions

No AWS or paid services are required.

Users

Admin

- Manage students
- Manage companies
- Approve placement drives
- View statistics
- Search users
- Blacklist companies or students

Company

- Register account
- Create profile
- Create placement drives
- View applicants
- Shortlist students
- Update application status

Student

- Register account
- Update profile
- Upload resume
- Browse drives
- Apply for drives
- View application history

Technology Stack

Backend

Java 25

Spring Boot

Spring MVC

Spring Data JPA

Hibernate

Spring Security

JWT

Validation

Swagger/OpenAPI

JUnit

Mockito

Database

MySQL Community Server

Alternative:

PostgreSQL

Frontend

React

Vite

Tailwind CSS

Axios

React Router DOM

React Hook Form

Caching

Redis

Messaging

RabbitMQ

Containerization

Docker

Docker Compose

Version Control

Git

GitHub

CI/CD

GitHub Actions

Folder Structure

Backend

src/main/java/com/saqib/placementportal

controller

service

repository

entity

dto

security

config

exception

util

Frontend

src

components

pages

layouts

hooks

context

services

utils

assets

Database Tables

User

Student

Company

PlacementDrive

Application

Resume

Notification

Role

Milestone 1

Spring Boot Fundamentals

Learn:

- Spring Boot
- Controller
- REST API

- Dependency Injection

Build:

Student CRUD APIs

GET

POST

PUT

DELETE

Milestone 2

Database and JPA

Learn:

- MySQL
- Entity
- Repository
- JPA
- Hibernate

Create tables:

Student

Company

Drive

Application

Use relationships:

OneToMany

ManyToOne

Milestone 3

Service Layer

Learn:

Service Architecture

DTO

ResponseEntity

Exception Handling

Validation

Build:

Business logic

Global exception handler

Custom exceptions

Milestone 4

Swagger API Documentation

Learn:

OpenAPI

Swagger UI

Test all APIs

Document endpoints

Milestone 5

Spring Security + JWT

Learn:

Authentication

Authorization

JWT

Password encryption

BCrypt

Roles:

ADMIN

COMPANY

STUDENT

Create:

Register API

Login API

JWT Filter

Protected APIs

Milestone 6

Placement Portal Backend

Student APIs

Company APIs

Drive APIs

Application APIs

Admin APIs

Features:

Apply drive

Prevent duplicate applications

Approve companies

Approve drives

Blacklist users

Search functionality

Pagination

Sorting

Milestone 7

React Frontend

Learn:

React

Vite

Tailwind CSS

React Router

Axios

Pages:

Login

Register

Admin Dashboard

Company Dashboard

Student Dashboard

Profile

Application History

Drive Details

Milestone 8

File Upload

Upload resumes

Store PDFs locally

Download resumes

Preview resumes

Learn:

MultipartFile

File Storage

Static Resources

Milestone 9

Redis Caching

Learn:

Redis

Spring Cache

Cacheable

Cache Evict

Cache:

Drives list

Company list

Statistics

Milestone 10

RabbitMQ

Learn:

Message Queues

Producer

Consumer

Background jobs

Features:

Email notifications

CSV export

Activity reports

Milestone 11

Docker

Learn:

Dockerfile

Docker Compose

Containerization

Run:

Spring Boot

MySQL

Redis

RabbitMQ

React

using docker-compose

Milestone 12

Testing

Learn:

JUnit

Mockito

Unit Testing

Integration Testing

Write tests for:

Service layer

Controller layer

Repository layer

Milestone 13

Git and GitHub

Branching

Commit strategy

Pull Requests

Version Control

Milestone 14

GitHub Actions

Automatic build

Automatic testing

CI pipeline

Milestone 15

Resume-Level Enhancements

Charts

Dashboard analytics

Admin statistics

Search and filters

Pagination

Dark mode

Responsive UI

Offer Letter PDF

Resume Analyzer

Email Notifications

CSV Export

Optional Features

OTP Login

Forgot Password

Google Login

Profile Picture Upload

WebSocket Notifications

Chat System

Interview Scheduler

Calendar Integration

AI Resume Analyzer

ATS Score Generator

Job Recommendation System

Docker Architecture

React

↓

Spring Boot

↓

MySQL

↓

Redis

↓

RabbitMQ

Everything runs locally

No cloud cost

Project Outcome

After completing this project, the following concepts will be covered:

Java

Spring Boot

Spring MVC

Dependency Injection

REST APIs

Hibernate

JPA

MySQL

Validation

DTO

Exception Handling

Swagger

Spring Security

JWT

React

Tailwind

Axios

File Upload

Redis

RabbitMQ

Docker

Testing

Git

GitHub Actions

CI/CD

Industry Project Structure

Resume Project Title

Placement Portal Application

Tech Stack:

Java 25, Spring Boot, Spring Security, JWT, Hibernate, JPA, MySQL, React, Tailwind CSS, Redis, RabbitMQ, Docker, Swagger, JUnit, GitHub Actions.

Built a full-stack role-based placement management system supporting Admin, Company and Student modules with secure JWT authentication, file upload, caching, background job processing and containerized deployment.